Curtis Chen

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Education

Cornell University – MEng Operations Research and Information Engineering **2022–2023**

- GPA: 3.9/4.0, Concentration in Machine Learning
- Relevant Coursework: Deep Learning, Applied ML, Linear Optimization, Causal Inference, Probability Theory

University College London – BS Physics **2019–2022**

- First Class Honors (GPA: 4.0/4.0)
- Men's 1st Team Basketball and Treasurer of the Basketball Club.

Work Experience

Data Analyst II, Costco Wholesale **March 2025 – Present**

- Fine-tuned a lightweight sentence transformer (PyTorch, HuggingFace) and built a batch inference pipeline orchestrated via Dagster and dbt to automate emission factor classification for new SKUs, reducing a 3-week manual process to under an hour and accelerating the corporate sustainability reporting cycle.
- Designed and deployed a text classification pipeline using TF-IDF and XGBoost, leveraging MLflow for experiment tracking and model versioning, to label 500,000+ historical SKUs by commodity ingredient and unlock previously inaccessible category-level sourcing insights.
- Built a suite of unsupervised deduplication and entity-resolution pipelines to validate vendor data at scale: address normalization via TF-IDF + cosine similarity, vendor deduplication using TF-IDF with DBSCAN clustering (scikit-learn), and disjoint-set merging for approximate record linkage — improving vendor data accuracy across Costco's supply chain and reducing downstream errors in emissions reporting.
- Rebuilt Costco's supply chain GHG emissions calculations as versioned Python/pandas pipelines, replacing ad hoc spreadsheets with reproducible, auditable workflows that became the standard for all subsequent emissions reporting.
- Led the migration of Costco's sustainability dataset from Google Sheets to BigQuery, creating a scalable data foundation and connecting dashboards to give 200+ stakeholders real-time visibility into supply chain emissions.
- Owned end-to-end supplier climate data collection across 800 vendors: designed Qualtrics surveys, extracted and transformed data using Python, and delivered segmentation analyses that classified suppliers by emissions maturity — directly informing Costco's tiering strategy and focusing outreach on the top 20% of vendors by emissions volume.

Projects

GPT-2 Architecture Implementation (Python, PyTorch)

- Implemented a complete GPT-2 transformer architecture from scratch following "Language Models are Unsupervised Multitask Learners," and added modern training optimizations such as mixed-precision training (bf16), FlashAttention, and PyTorch JIT compilation, achieving 10x faster training throughput vs. a naive baseline.
- Replicated model performance benchmarks on the HellaSwag dataset using hyperparameters from GPT-2 and GPT-3 papers, achieved ~30% normalized accuracy on HellaSwag, matching GPT-2's reported benchmark.

PaliGemma Architecture Implementation (Python, PyTorch)

- Implemented a multimodal vision-language model from scratch, including the SigLIP image encoder and Gemma decoder, integrating modern attention modifications like RoPE, KV-cache, and grouped query attention.
- Evaluated model outputs on VQAv2 and TextVQA benchmarks, achieving 83% and 55% respectively at 224px, validating alignment with the original PaliGemma paper's reported performance.

Skills

- **ML Libraries:** PyTorch, scikit-learn, XGBoost, LightGBM, MLflow, HuggingFace Transformers
- **Cloud/Databases:** BigQuery, Postgres, DuckDB, PySpark, dbt, Dagster
- **Data/Visualization Libraries:** pandas, numpy, scipy, statsmodels, matplotlib, Streamlit
- **Tools:** PowerBI, Looker, Gurobi
- **Languages:** Python, SQL